ChatGPT said:

Causal discovery in time series: Practical considerations for PC, PCMCI, Granger causality, LiNGAM, NOTEARS, and GES

The key point is that **causal discovery algorithms do not all make the same assumptions about data, and preprocessing that helps one algorithm can harm another**. For example, PCMCI is designed for time-series causal discovery with autocorrelation and lagged effects, whereas standard LiNGAM, NOTEARS, and GES are primarily formulated for learning causal graphs from observational data represented as independent samples. Time-series adaptations exist for several of these methods.

The answers below distinguish general principles from algorithm-specific behavior. They are intended as practical guidance rather than universal guarantees, since results depend on sample size, causal effect strength, graph sparsity, noise, the conditional-independence test, and the underlying data-generating process.

Science Adv… +3

1. Do we need to standardize or normalize time series before applying causal discovery algorithms? What happens if we do not?

Standardization is often useful, but it is not universally required. **Standardization** typically means subtracting the mean and dividing by the standard deviation; **normalization** may mean rescaling values to a fixed range. These operations are not interchangeable with detrending, differencing, or removing seasonality.

For PC and PCMCI, standardization is generally unnecessary when using scale-invariant conditional-independence tests such as partial correlation. However, it can improve numerical stability and make regularization parameters easier to interpret. For Granger causality, rescaling variables generally does not change ideal linear-model significance tests, although poor numerical conditioning can affect estimation.

For LiNGAM, invertible affine rescaling ordinarily preserves the underlying causal structure, but distributional assumptions must still hold. For NOTEARS, which uses a penalized optimization objective, variable scales can substantially affect the balance between fit and regularization. GES also depends on its scoring function and any regularization.

Without standardization, large-scale variables may dominate scale-sensitive penalties or optimization, and small-scale effects may be harder to estimate numerically. Conversely, nonlinear normalization can distort relationships. **Standardize using training-period statistics when appropriate, and do not mistake standardization for making a nonstationary series stationary.**

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2. How many input data points are needed to obtain statistically significant results?

There is no universal minimum sample size for causal discovery. The required number depends on the number of variables, graph sparsity, effect sizes, noise, autocorrelation, maximum lag, number of conditioning variables, and the desired statistical power. Statistical significance for an individual edge is also different from reliably recovering the entire causal graph.

For time-series methods, the effective sample size may be much smaller than the raw number of observations because adjacent points are autocorrelated. PCMCI can perform well with hundreds of observations in some sparse, moderately dimensional settings, but this is not a guarantee. Strong autocorrelation, weak causal effects, nonlinear relationships, and many candidate edges can require substantially more data. Granger causality needs enough observations to estimate its autoregressive model reliably. PC and GES can become sample-hungry when conditioning sets grow, while NOTEARS and LiNGAM also need sufficient observations relative to model complexity and their assumptions.

Practical starting point: benchmark sample sizes such as 200, 500, 1,000, and 5,000 observations where feasible, using synthetic data that resembles your application. Evaluate edge precision, recall, and stability across repeated simulations or bootstrap samples. Use domain-specific simulations and power analysis to select the sample size rather than treating any threshold as universal.

Science Adv… +2

3. Does performance break down with a large number of features? What is the expected threshold?

Yes, performance can degrade as the number of variables increases, but there is no fixed feature-count threshold. The important quantities are the number of observations, the number of candidate edges, the density of the true graph, and the complexity of the conditional-independence tests or optimization problem.

PC can become computationally expensive when many variables must be considered in large conditioning sets. Standard multivariate Granger causality can suffer when the number of model parameters grows relative to the number of observations. PCMCI addresses this problem through parent-selection and momentary conditional-independence testing, and published experiments demonstrate useful performance with dozens to hundreds of time series under suitable conditions. This does not mean every dataset of that size will be reliable.

Science Adv… +2

LiNGAM and GES can also become expensive as the graph search space grows. NOTEARS avoids discrete graph search through continuous optimization, but its optimization cost and statistical difficulty can still increase considerably with dimensionality.

As a rough engineering guideline, begin with tens of variables, validate performance, and then scale to hundreds or more if justified. These are experimental milestones, not hard limits. For large datasets, exploit sparsity, restrict plausible lags, use domain knowledge, and benchmark against known or simulated ground truth.

4. How do these algorithms handle multicollinearity?

Multicollinearity occurs when two or more variables are strongly correlated or contain overlapping information. It is particularly problematic when causal discovery must distinguish direct effects from indirect effects or shared causes.

For linear Granger causality, highly correlated predictors can make autoregressive coefficients unstable, inflate their standard errors, and reduce power to identify individual causal links. PC and PCMCI use conditional-independence tests; strong dependencies between conditioning variables can make these tests unreliable or leave multiple causal structures statistically indistinguishable. PCMCI's parent-selection step reduces unnecessary conditioning, but it cannot eliminate the fundamental identifiability problem.

Science Adv… +1

LiNGAM can identify directions through independent, non-Gaussian disturbances under its assumptions, but near-collinearity can make estimation numerically unstable. NOTEARS may select one among several highly correlated predictors, depending on the loss, regularization, and sample. GES may similarly favor one graph among alternatives with comparable scores.

Perfect collinearity can make individual causal effects unidentifiable from observational data alone. Even strong but imperfect collinearity can create unstable edges across samples.

Recommended practice: inspect correlations and condition numbers, check edge stability across resamples, and use domain knowledge or interventions where possible. Regularization can stabilize estimation, but it does not automatically establish which correlated variable is the true cause.

5. How do they handle nonstationarity?

Nonstationarity means that important statistical properties of a series, such as its mean, variance, or dependence structure, change over time. It can arise from trends, seasonality, changing operating conditions, regime switches, or evolving causal mechanisms.

Classical Granger causality commonly assumes a stable autoregressive process. Trends or unit roots can produce misleading relationships unless addressed appropriately, for example through differencing, detrending, cointegration analysis, or a suitable time-varying model. PC and standard PCMCI also rely on assumptions about the underlying distribution and causal structure; PCMCI has shown robustness to some trends and other nonstationarities, but it is not immune to arbitrary regime changes.

Science Adv… +2

Standard LiNGAM and GES generally assume that the samples represent a stable underlying distribution. Standard NOTEARS has a similar issue unless its model explicitly accommodates time variation. Time-series-specific variants may incorporate temporal structure, but that capability should be verified for the particular implementation.

Preprocessing should follow diagnosis: remove known seasonality or deterministic trends when appropriate, consider differencing or cointegration for integrated processes, or use rolling-window, regime-specific, or time-varying causal discovery when the graph itself may evolve.

Importantly, **forcing every series to be stationary is not always the correct solution**. Differencing may remove meaningful long-run relationships. The choice of method should reflect whether the nonstationarity is in the observed level, the noise distribution, or the causal mechanism.

6. Can these algorithms handle cyclic graphs?

The answer depends on whether the cycle is contemporaneous or involves different time points.

Standard PC, standard GES, standard LiNGAM, and the original NOTEARS formulation target directed acyclic graphs (DAGs). They cannot represent arbitrary directed cycles in the graph they are designed to learn. NOTEARS explicitly enforces acyclicity, and standard LiNGAM assumes a linear acyclic causal ordering.

Salesforce … +1

Granger causality and PCMCI can represent feedback over time. For example, $X_t \to Y_{t+1}$ and $Y_t \to X_{t+1}$ form a feedback loop in the time-series process, even though the corresponding time-unrolled graph is acyclic. This is a natural way to represent dynamic feedback without requiring a cyclic graph among variables at a single time point.

Contemporaneous cycles are more difficult. A model in which X_t directly causes Y_t and Y_t directly causes X_t may require a simultaneous-equations model or a specialized cyclic causal-discovery method. PCMCI+ can discover contemporaneous links under its assumptions, but it does not generally solve unrestricted cyclic structural models.

Recommendation: if feedback is expected, clarify whether it occurs with a measurable time delay. For lagged feedback, use a time-series method; for instantaneous feedback, investigate methods explicitly designed for cyclic structural causal models.

7. What if there is a change point, or the distribution of a causal variable changes?

A change point can alter the mean, variance, noise distribution, causal strength, or the graph itself. These are distinct situations. Suppose X causes Y, and the distribution of X changes after a particular time. The causal relationship may remain unchanged, or it may change too. A distribution shift alone does not establish that the causal edge has changed.

If the data from both regimes are pooled, standard PC, PCMCI, Granger causality, LiNGAM, NOTEARS, or GES may estimate a graph that describes neither regime accurately. A change in variance can alter test power; a change in the relationship between X and Y can yield different edge strengths or directions. A change in an unobserved common cause can also create apparent new relationships.

A practical strategy is to detect candidate change points, estimate graphs separately within sufficiently large segments, and compare edge presence and causal strength across segments. Rolling-window discovery can track gradual evolution, although short windows increase uncertainty. If the causal mechanism is stable but the input distribution shifts, invariance-based approaches can help distinguish stable mechanisms from regime-specific associations.

Key caution: do not interpret a changed graph as evidence of a changed causal mechanism until you have considered sample-size differences, measurement changes, confounding, and changes in statistical power.

8. For high-frequency time series, how frequently should we sample to preserve causal structure while reducing the number of observations?

There is no universal sampling frequency that preserves causal structure. The appropriate frequency depends on the timescale of the underlying dynamics, the shortest causal delay of interest, measurement noise, and whether the process is continuous-time or discrete-time.

If sampling is too slow, a cause and its effect may occur between observations, making their temporal order impossible to determine. Different causal pathways can become observationally indistinguishable, and rapid effects may appear instantaneous. If sampling is excessively fast, the dataset may contain many redundant, highly autocorrelated observations, increasing computational cost without providing proportionally more independent information. Granger causality, in particular, assumes a discrete, regularly sampled process and can lose identifiability when the sampling interval is poorly matched to the true causal delays.

PMC +1

A sensible approach is to estimate the relevant physical or operational timescales, examine autocorrelation and spectral characteristics, and test multiple sampling intervals. Downsampling should generally be preceded by appropriate anti-alias filtering when high-frequency components could alias into lower frequencies. Filtering itself must be chosen carefully because it can introduce phase shifts or distort temporal relationships.

Compare discovered graphs across candidate sampling rates using synthetic data, known events, and edge stability. Preserve the original data for validation, and do not assume that a graph learned at one frequency will remain equivalent at another. For continuous-time systems, consider methods explicitly designed for continuous-time causal discovery rather than relying solely on discrete-time algorithms.

Cross-algorithm summary

The table below summarizes the main practical differences.

Algorithm	Scaling sensitivity	High-dimensional considerations	Cycles and temporal structure
PC	Depends on conditional-independence test	Can become expensive with large conditioning sets	Standard version assumes a DAG
PCMCI	Usually low with scale-invariant tests	Designed for sparse, high-dimensional time series	Supports lagged feedback; PCMCI+ also handles contemporaneous links under its assumptions
Granger causality	Usually low for well-specified linear tests	Multivariate regression can become parameter-heavy	Naturally captures lagged feedback
LiNGAM	Scaling usually preserves structure	Estimation and identifiability become harder as dimension grows	Standard version assumes a linear DAG with non-Gaussian independent errors

NOTEARS	Potentially high due to optimization and regularization	Continuous optimization can be costly at large dimensions	Standard formulation explicitly enforces acyclicity
GES	Depends on the score and search configuration	Graph search becomes expensive as dimension grows	Standard version searches over DAG equivalence classes

These distinctions follow from the algorithms' standard formulations; specialized variants may have different capabilities.

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Recommended experimental protocol

For a real-world causal discovery project, I would prioritize the following checks before selecting a final algorithm:

1. **Data preparation:** inspect missingness, sampling regularity, trends, seasonality, outliers, scaling, and potential change points.
2. **Temporal specification:** establish the maximum causal lag and assess whether contemporaneous effects or feedback are plausible.
3. **Benchmarking:** simulate data with known causal graphs and realistic autocorrelation, noise, dimensionality, and regime changes.
4. **Robustness:** vary sample size, sampling frequency, preprocessing choices, and algorithm hyperparameters. Measure edge precision, recall, and graph stability.
5. **Statistical validation:** account for multiple edge tests, uncertainty, and the distinction between predictive relationships and causal identification.
6. **Scientific validation:** use domain knowledge, interventions, or independent datasets to assess whether the discovered edges are plausible.

The most important practical principle is that **a statistically significant edge is not, by itself, proof of causality**. Causal interpretation depends on the algorithm's assumptions, adequate data, correct temporal specification, and the absence or appropriate handling of confounding.